

502 Bad Gateway after the first deploy to the VM

Team Mahokshahvata · GCP project `project-262a2026-879b-4b49-88e` · Incident: Oct 1, 2026 (first deploy)
VM `mahokshahvata` : e2-micro (shared vCPU, 1 GB RAM), us-central1-a, Ubuntu 24.04 · FastAPI app (uvicorn, Python 3.14) on `127.0.0.1:8000` under `systemd`, behind Caddy on port 80

After our first deploy, the site returned **502 Bad Gateway** for about 90 seconds. The app was healthy but still starting, and our deploy script reported success too early. We fixed the script to wait until the app answers.

Symptom: exact error

The deploy script printed `Deployed: https://34.66.1.208.sslip.io`, but the browser showed **HTTP ERROR 502 (Bad Gateway)**. Every request failed the same way:

```
$ curl -s -o /dev/null -w "%{http_code}\n" https://34.66.1.208.sslip.io/
502
```

The same code answered on `localhost:8000` on the laptop immediately. (That first VM was later recreated; the app now runs at `http://34.122.164.209`.)

Diagnosis

#	COMMAND / CHECK	WHAT IT SHOWED	SO...
1	Error type	502, not a timeout	Request reached the VM and Caddy; firewall was fine. Caddy couldn't reach the app.
2	<code>systemctl is-active mahokshahvata caddy</code>	<code>active / active</code>	The app had not crashed.
3	<code>sudo journalctl -u mahokshahvata</code>	<code>Started mahokshahvata.service, but no Application startup complete.</code>	Process existed but wasn't listening yet.
4	<code>free -m</code>	953 MB total, ~480 MB available, swap unused	Not out of memory.
5	<code>curl</code> every 15 s	<code>502, 502, 502, 200</code>	App came up by itself after ~60–90 s.

Root cause

Not a crash and not the network. Before uvicorn starts listening, the app imports `pandas` and `yfinance`, loads the 3,000-stock market snapshot (`momentum/market_cache.json`) and fills in sectors. That takes seconds on a laptop but **60–90 seconds** on the e2-micro's shared vCPU. The script printed "Deployed" as soon as `systemctl restart` returned, so during that window Caddy had nothing to forward to and returned 502.

Fix: exact change

We changed `deploy/gcp.sh` so it reports success only once the app answers through Caddy. After restarting the service, it polls for up to 3 minutes and fails loudly if the app never comes up:

```
# Startup takes ~60–90s on an e2-micro; Caddy returns 502 until uvicorn listens.
for i in $(seq 60); do curl -sf -o /dev/null localhost:80/ && exit 0; sleep 3; done
echo 'App did not answer within 3 minutes; check journalctl -u mahokshahvata' >&2; exit 1
```

Result: the URL works the moment the script prints `Deployed:` , and a real startup failure (e.g. a missing dependency) now shows up as a failed deploy instead of a silent 502.

Lesson: "the service is active" is not "the service is ready". Check readiness with a real HTTP request, not just the process state.